SPI Slave CDC Bridge — E. Connectivity / Hierarchy / Port Tables

RTL: spi_bridge_top.v (1,293 lines). 정적 검토. NC: u_spi_master.o_rx_captured, u_spi_master.o_rx_captured_valid, u_tx_fifo.o_rd_level

Hierarchy

Module	Instance	Parent	Clock	Reset	Function	구분
spi_bridge_top	(top)	—	ACLK (입력), SCLK (내부 생성)	ARESETn (async 입력)	최상위: 클럭/리셋 생성, 내부 SPI master↔slave 직결, 두 개의 async FIFO, 레지스터 블록, busy 2FF 동기화, debug 출력	top
reset_sync	u_reset_sync_aclk	spi_bridge_top	ACLK	ARESETn → aclk_rst_n	async assert / sync release (2FF), ACLK 도메인 리셋	인스턴스
reset_sync	u_reset_sync_sclk	spi_bridge_top	SCLK	ARESETn → sclk_rst_n	async assert / sync release (2FF), SCLK 도메인 리셋	인스턴스
sclk_gen	u_sclk_gen	spi_bridge_top	ACLK	aclk_rst_n (async)	ACLK÷4 분주 (r_toggle_phase + o_sclk), 25 MHz	인스턴스
BUFG (Xilinx primitive)	u_bufg_sclk	spi_bridge_top	—	—	w_sclk_raw → SCLK 전역 클럭 버퍼 (RTL 모듈 아님, 라이브러리 primitive)	인스턴스
spi_master	u_spi_master	spi_bridge_top	SCLK	sclk_rst_n (async)	내부 SPI master: 4-state FSM, 8-bit 프레임 송신 (패턴 r_tx_byte++), MISO 수신	인스턴스
spi_slave	u_spi_slave	spi_bridge_top	SCLK	sclk_rst_n (async)	SPI slave: 4-state FSM, RX shift, TX hold/shift, FIFO 인터페이스	인스턴스
async_fifo	u_rx_fifo	spi_bridge_top	wr: SCLK / rd: ACLK	wr: sclk_rst_n / rd: aclk_rst_n	RX FIFO, DATA_WIDTH=8, ADDR_WIDTH=4 (16×8), SCLK→ACLK	인스턴스
async_fifo	u_tx_fifo	spi_bridge_top	wr: ACLK / rd: SCLK	wr: aclk_rst_n / rd: sclk_rst_n	TX FIFO, DATA_WIDTH=8, ADDR_WIDTH=2 (4×8), ACLK→SCLK	인스턴스
pulse_sync	u_overflow_pulse_sync	spi_bridge_top	src: SCLK / dst: ACLK	src: sclk_rst_n / dst: aclk_rst_n	RX overflow pulse SCLK→ACLK (toggle + 2FF + XOR)	인스턴스
register_block	u_register_block	spi_bridge_top	ACLK	aclk_rst_n (sync)	레지스터 버스(0x00..0x10), RX prefetch 버퍼, drain, overflow sticky, TX write	인스턴스

Connectivity (64)

Source Instance.Port	Signal	Width	Destination Instance.Port	Clock Domain
(top).ACLK	ACLK	1	u_reset_sync_aclk.clk	ACLK/async
(top).ACLK	ACLK	1	u_sclk_gen.clk	ACLK/async
(top).ACLK	ACLK	1	u_rx_fifo.rd_clk	ACLK/async
(top).ACLK	ACLK	1	u_tx_fifo.wr_clk	ACLK/async
(top).ACLK	ACLK	1	u_overflow_pulse_sync.dst_clk	ACLK/async
(top).ACLK	ACLK	1	u_register_block.ACLK	ACLK/async
(top).ARESETn	ARESETn	1	u_reset_sync_aclk.arst_n	ACLK/async
(top).ARESETn	ARESETn	1	u_reset_sync_sclk.arst_n	ACLK/async
(top).i_bus_addr	i_bus_addr	8	u_register_block.i_bus_addr	ACLK
(top).i_bus_wr_en	i_bus_wr_en	1	u_register_block.i_bus_wr_en	ACLK
(top).i_bus_wdata	i_bus_wdata	8	u_register_block.i_bus_wdata	ACLK
(top).i_bus_rd_en	i_bus_rd_en	1	u_register_block.i_bus_rd_en	ACLK
u_register_block.o_bus_rdata	o_bus_rdata	8	(top).o_bus_rdata	ACLK
u_register_block.o_overflow	o_overflow	1	(top).o_overflow	ACLK
u_register_block.o_rx_valid	o_rx_valid	1	(top).o_rx_valid	ACLK
u_reset_sync_aclk.rst_n_sync	aclk_rst_n	1	u_sclk_gen.rst_n	ACLK
u_reset_sync_aclk.rst_n_sync	aclk_rst_n	1	u_rx_fifo.rd_rst_n	ACLK
u_reset_sync_aclk.rst_n_sync	aclk_rst_n	1	u_tx_fifo.wr_rst_n	ACLK
u_reset_sync_aclk.rst_n_sync	aclk_rst_n	1	u_overflow_pulse_sync.dst_rst_n	ACLK
u_reset_sync_aclk.rst_n_sync	aclk_rst_n	1	u_register_block.rst_n	ACLK

u_bufg_sclk.O	SCLK	1	u_reset_sync_sclk.clk	SCLK(생성)
u_bufg_sclk.O	SCLK	1	u_spi_master.SCLK	SCLK(생성)
u_bufg_sclk.O	SCLK	1	u_spi_slave.SCLK	SCLK(생성)
u_bufg_sclk.O	SCLK	1	u_rx_fifo.wr_clk	SCLK(생성)
u_bufg_sclk.O	SCLK	1	u_tx_fifo.rd_clk	SCLK(생성)
u_bufg_sclk.O	SCLK	1	u_overflow_pulse_sync.src_clk	SCLK(생성)
u_bufg_sclk.O	SCLK	1	(top).o_dbg_sclk[assign]	SCLK(생성)
u_reset_sync_sclk.rst_n_sync	sclk_rst_n	1	u_spi_master.rst_n	SCLK
u_reset_sync_sclk.rst_n_sync	sclk_rst_n	1	u_spi_slave.rst_n	SCLK
u_reset_sync_sclk.rst_n_sync	sclk_rst_n	1	u_rx_fifo.wr_rst_n	SCLK
u_reset_sync_sclk.rst_n_sync	sclk_rst_n	1	u_tx_fifo.rd_rst_n	SCLK
u_reset_sync_sclk.rst_n_sync	sclk_rst_n	1	u_overflow_pulse_sync.src_rst_n	SCLK
u_sclk_gen.o_sclk	w_sclk_raw	1	u_bufg_sclk.I	SCLK(생성)
u_spi_master.o_cs_n	w_cs_n	1	u_spi_slave.i_cs_n	SCLK
u_spi_master.o_cs_n	w_cs_n	1	(top).o_dbg_cs_n[assign]	SCLK
u_spi_master.o_mosi	w_mosi	1	u_spi_slave.i_mosi	SCLK
u_spi_master.o_mosi	w_mosi	1	(top).o_dbg_mosi[assign]	SCLK
u_spi_slave.o_miso	w_miso	1	u_spi_master.i_miso	SCLK
u_spi_slave.o_miso	w_miso	1	(top).o_dbg_miso[assign]	SCLK
u_spi_master.o_busy	w_spi_busy_master	1	top.w_spi_busy(OR)	SCLK→ACLK (OR+2FF)
u_spi_slave.o_rx_wr_en	w_rx_wr_en	1	u_rx_fifo.i_wr_en	SCLK
u_spi_slave.o_rx_wr_data	w_rx_wr_data	8	u_rx_fifo.i_wr_data	SCLK
u_rx_fifo.o_full	w_rx_fifo_full	1	u_spi_slave.i_rx_fifo_full	SCLK→(CDC-01: ACLK 직결)
u_rx_fifo.o_full	w_rx_fifo_full	1	u_register_block.i_rx_fifo_full	SCLK→(CDC-01: ACLK 직결)
u_spi_slave.o_rx_overflow_pulse	w_rx_overflow_pulse	1	u_overflow_pulse_sync.i_src_pulse	SCLK→ACLK (pulse_sync)
u_spi_slave.o_tx_fifo_rd_en	w_tx_fifo_rd_en	1	u_tx_fifo.i_rd_en	SCLK
u_tx_fifo.o_rd_data	w_tx_fifo_rd_data	8	u_spi_slave.i_tx_fifo_rd_data	SCLK
u_tx_fifo.o_empty	w_tx_fifo_empty	1	u_spi_slave.i_tx_fifo_empty	SCLK
u_spi_slave.o_spi_busy	w_spi_busy_slave	1	top.w_spi_busy(OR)	SCLK→ACLK (OR+2FF)
u_register_block.o_rx_fifo_rd_en	w_rx_fifo_rd_en	1	u_rx_fifo.i_rd_en	ACLK
u_rx_fifo.o_rd_data	w_rx_fifo_rd_data	8	u_register_block.i_rx_fifo_rd_data	ACLK
u_rx_fifo.o_empty	w_rx_fifo_empty	1	u_register_block.i_rx_fifo_empty	ACLK
u_rx_fifo.o_rd_level	w_rx_fifo_level	5	u_register_block.i_rx_fifo_level	ACLK
u_register_block.o_tx_fifo_wr_en	w_tx_fifo_wr_en	1	u_tx_fifo.i_wr_en	ACLK
u_register_block.o_tx_fifo_wr_data	w_tx_fifo_wr_data	8	u_tx_fifo.i_wr_data	ACLK
u_tx_fifo.o_full	w_tx_fifo_full	1	u_register_block.i_tx_fifo_full	ACLK
u_overflow_pulse_sync.o_dst_pulse	w_rx_overflow_pulse_sync	1	u_register_block.i_rx_overflow_pulse_sync	ACLK
top.r_spi_busy_sync(2FF reg)	r_spi_busy_sync	1	u_register_block.i_spi_busy_sync	ACLK
(top) assign	SCLK	1	(top).o_dbg_sclk	SCLK(생성)
(top) assign	w_cs_n	1	(top).o_dbg_cs_n	SCLK
(top) assign	w_mosi	1	(top).o_dbg_mosi	SCLK
(top) assign	w_miso	1	(top).o_dbg_miso	SCLK
top.w_spi_busy(OR)	w_spi_busy (= w_spi_busy_master || w_spi_busy_slave)	1	top.r_spi_busy_meta (FF D)	SCLK→ACLK 경계 (CDC-02: OR 조합 후 2FF)
top.r_spi_busy_meta	r_spi_busy_meta	1	top.r_spi_busy_sync (FF D)	ACLK

spi_bridge_top ports

Dir	Width	Port
input	1	ACLK
input	1	ARESETn

input	[7:0]	i_bus_addr
input	1	i_bus_wr_en
input	[7:0]	i_bus_wdata
input	1	i_bus_rd_en
output	[7:0]	o_bus_rdata
output	1	o_overflow
output	1	o_rx_valid
output	1	o_dbg_sclk
output	1	o_dbg_cs_n
output	1	o_dbg_mosi
output	1	o_dbg_miso

spi_master ports

Dir	Width	Port
input	1	SCLK
input	1	rst_n
output	1	o_cs_n
output	1	o_mosi
input	1	i_miso
output	[7:0]	o_rx_captured
output	1	o_rx_captured_valid
output	1	o_busy

spi_slave ports

Dir	Width	Port
input	1	SCLK
input	1	rst_n
input	1	i_cs_n
input	1	i_mosi
output	1	o_miso
output	1	o_rx_wr_en
output	[7:0]	o_rx_wr_data
input	1	i_rx_fifo_full
output	1	o_rx_overflow_pulse
output	1	o_tx_fifo_rd_en
input	[7:0]	i_tx_fifo_rd_data
input	1	i_tx_fifo_empty
output	1	o_spi_busy

async_fifo ports

Dir	Width	Port
input	1	wr_clk
input	1	wr_rst_n
input	1	i_wr_en
input	[DATA_WIDTH-1:0]	i_wr_data
output	1	o_full
input	1	rd_clk
input	1	rd_rst_n
input	1	i_rd_en
output	[DATA_WIDTH-1:0]	o_rd_data

output	1	o_empty
output	[ADDR_WIDTH:0]	o_rd_level

register_block ports

Dir	Width	Port
input	1	ACLK
input	1	rst_n
input	[7:0]	i_bus_addr
input	1	i_bus_wr_en
input	[7:0]	i_bus_wdata
input	1	i_bus_rd_en
output	[7:0]	o_bus_rdata
output	1	o_rx_fifo_rd_en
input	[7:0]	i_rx_fifo_rd_data
input	1	i_rx_fifo_empty
input	1	i_rx_fifo_full
input	[4:0]	i_rx_fifo_level
output	1	o_tx_fifo_wr_en
output	[7:0]	o_tx_fifo_wr_data
input	1	i_tx_fifo_full
input	1	i_spi_busy_sync
input	1	i_rx_overflow_pulse_sync
output	1	o_overflow
output	1	o_rx_valid

pulse_sync ports

Dir	Width	Port
input	1	src_clk
input	1	src_rst_n
input	1	i_src_pulse
input	1	dst_clk
input	1	dst_rst_n
output	1	o_dst_pulse

reset_sync ports

Dir	Width	Port
input	1	clk
input	1	arst_n
output	1	rst_n_sync

sclk_gen ports

Dir	Width	Port
input	1	clk
input	1	rst_n
output	1	o_sclk

BUFG ports

Dir	Width	Port
input	1	I
output	1	O